

Posterior-Based Decision Stability in Urban Flood Risk Prioritisation

Mikel Martinez Mugica

March 2026

Abstract

Urban risk assessments often produce ranked lists of assets or locations to guide infrastructure investment and resilience planning. These rankings are typically derived from deterministic risk scores or expected impact estimates. While uncertainty may be quantified within individual model components, it is rarely propagated through the full exposure–hazard–impact chain to the prioritisation decisions themselves.

This paper introduces a reproducible Bayesian baseline for propagating uncertainty from exposure and hazard proxies to decision-relevant prioritisation metrics in urban pluvial risk assessment.

Using Rotterdam as a case study (~221,324 buildings), we construct an exposure–hazard–outcome pipeline and perform Bayesian inference to estimate posterior distributions of asset-level impact probabilities. Prioritisation decisions are then derived from posterior draws rather than point estimates.

We quantify decision stability using posterior-derived metrics including top-k membership probability and borderline share. Results suggest that most assets exhibit stable prioritisation outcomes under the current modelling assumptions, while a small fraction occupy an unstable decision boundary where ranking uncertainty is concentrated.

This work provides a minimal probabilistic baseline for analysing decision stability in urban risk prioritisation and establishes a foundation for future extensions incorporating hazard uncertainty and cross-city transfer. The outcome variable is synthetic and used solely to validate the probabilistic inference pipeline.

1. Introduction

Urban resilience planning frequently relies on ranked lists of assets requiring intervention, such as buildings, infrastructure segments, or neighbourhoods. These rankings typically emerge from deterministic risk scores or expected damage estimates.

However, risk estimation involves substantial uncertainty arising from:

- hazard variability
- exposure measurement error
- model specification
- limited observational data

Although uncertainty may be analysed within individual modelling components, it is rarely propagated through the full modelling chain to the final prioritisation decisions.

As a result, the stability of ranked intervention lists remains largely unexplored.

This paper proposes treating **prioritisation stability itself as the primary inferential object**. While uncertainty propagation has been widely studied within individual components of flood risk models, particu-

larly in Bayesian flood damage modelling and hydrological ensemble simulations (Sairam et al., 2019; Mohor et al., 2021; McMillan & Brasington, 2008), the stability of the resulting prioritisation decisions has received comparatively little attention.

Instead of asking only:

Which assets appear most at risk?

we ask:

How stable are prioritisation decisions when uncertainty is propagated through the modelling pipeline?

To explore this question, we construct a reproducible Bayesian baseline for urban pluvial risk assessment and derive prioritisation metrics directly from posterior distributions.

The contribution of this paper is therefore methodological: rather than improving hazard or damage estimation alone, the framework evaluates how uncertainty propagates into the stability of prioritisation decisions themselves.

Phase-1 Contributions

This paper provides:

- A reproducible **exposure–hazard–outcome modelling pipeline**
- A **Bayesian inference baseline** for asset-level impact probability
- **Posterior-derived prioritisation metrics**
- A **citywide decision stability analysis** for Rotterdam (~221k assets)

Future work will extend this framework to structural uncertainty in hazard representation and cross-city transfer.

2. Conceptual Framework

Urban flood risk assessments typically follow a conceptual structure linking:

Exposure → Hazard → Impact (Outcome)

where exposure describes asset characteristics, hazard represents environmental forcing, and impact represents realised damage or disruption.

Probabilistic approaches to flood risk modelling have previously been applied to individual components of this chain, including Bayesian damage estimation (Sairam et al., 2019; Mohor et al., 2021) and probabilistic flood susceptibility models using Bayesian networks (Wu et al., 2019, 2020).

In probabilistic terms this can be expressed as a generative model:

Data → Exposure → Hazard → Outcome ↓ Posterior inference ↓ Decision metrics

Posterior inference produces distributions over model parameters and asset-level risk probabilities. Rather than collapsing these distributions to point estimates, we derive decision quantities directly from the posterior.

This allows prioritisation decisions to be analysed probabilistically.

Graphical model

Figure 1 illustrates the hierarchical exposure–hazard–impact structure underlying the model.

Figure 1 — Hierarchical Exposure–Hazard–Impact Model (general formulation)

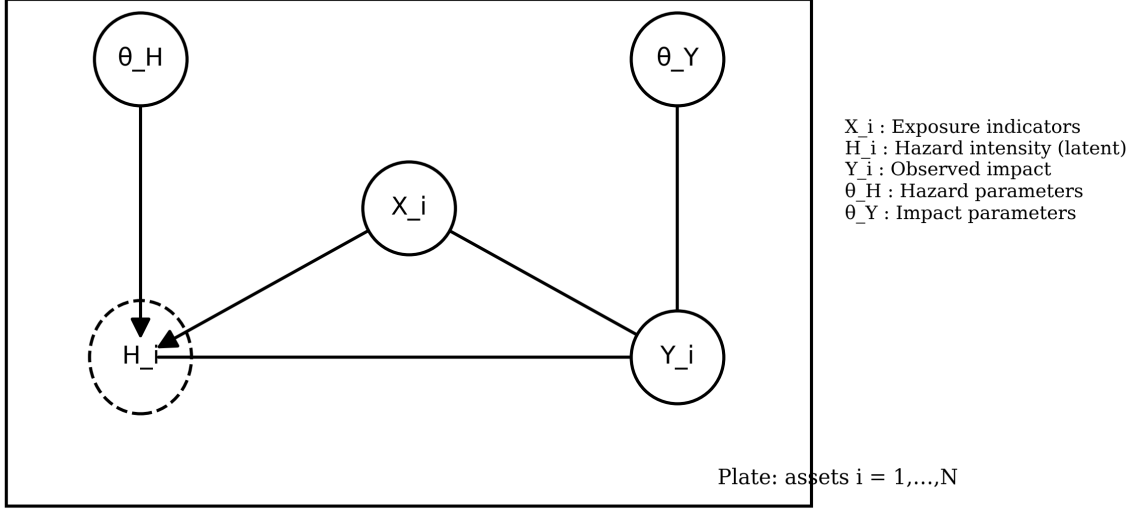


Figure 1: Hierarchical exposure–hazard–impact model

Decision stability concept

Figure 2 illustrates how posterior uncertainty propagates to asset ranking distributions and top-k membership probabilities.

These quantities form the basis for the decision stability metrics analysed in the Rotterdam case study.

Domain-shift stress test (conceptual)

Figure 3 illustrates the conceptual domain-shift experiment proposed for later phases of the research. The model specification remains fixed while the framework is applied across different cities. If the generative assumptions are stable, posterior ranking behaviour should remain similar across cities. Under domain shift, ranking variability is expected to increase, producing wider posterior rank distributions and larger decision instability.

This experiment forms the basis of the cross-city transfer analysis planned for future phases of the research.

3. Data and Study Area

The modelling pipeline is applied to the municipality of **Rotterdam, Netherlands**, containing approximately **221,324 buildings**.

The dataset integrates several open data sources:

- **OpenStreetMap building footprints**
- **Dutch hydrography datasets**
- **ERA5-Land precipitation data**

Exposure proxy

A deterministic exposure proxy $\hat{E}_i$ is constructed using spatial indicators of building proximity to water bodies and local hydrographic density.

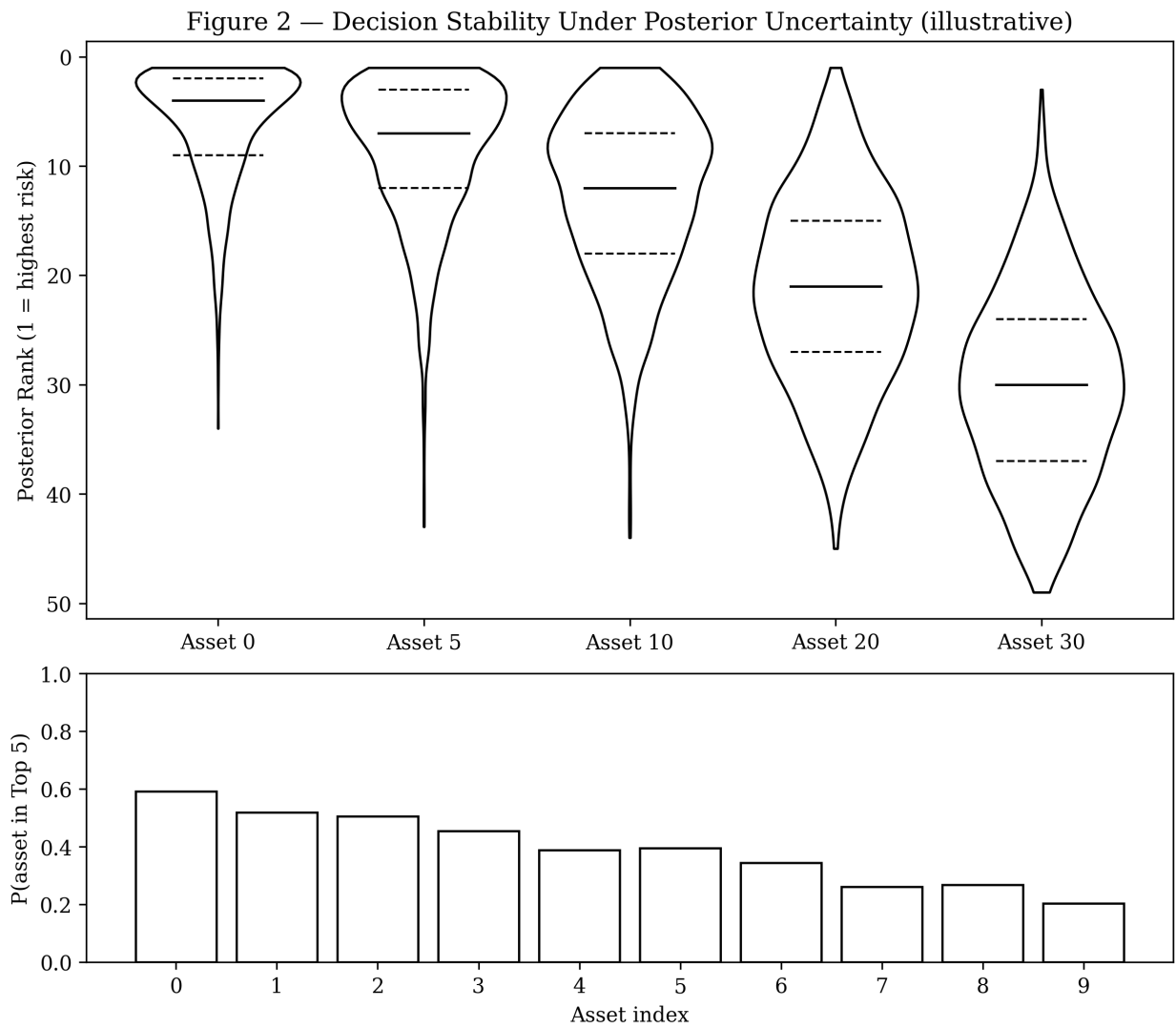


Figure 2: Decision stability concept

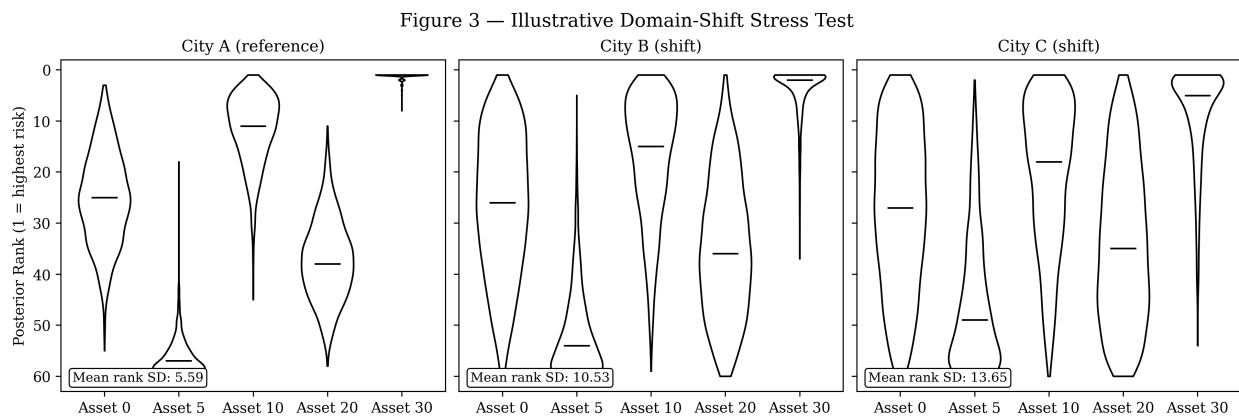


Figure 3: Domain-shift stress test concept

In Phase-1, exposure is represented by a deterministic index constructed from hydrography proximity features (distance to nearest water body and water length density within multiple buffer radii) and standardized prior to modelling.

Hazard proxy

A pluvial hazard proxy (H_i) is derived from precipitation forcing indicators. The hazard representation is intentionally simplified in Phase-1 to focus on validating the inference pipeline.

Hazard is represented by a deterministic pluvial forcing proxy derived from ERA5-Land precipitation indicators and standardized prior to modelling.

Outcome variable

The outcome variable represents a **synthetic damage indicator**, used solely to validate the probabilistic inference structure. No claims of empirical calibration are made in this phase.

4. Bayesian Model

The model defines a simple generative structure linking exposure, hazard proxy, and binary impact outcome. Asset-level impact probability is modelled using a logistic regression structure:

$$p_i = \text{logit}^{-1}(\alpha + \beta_E E_i + \beta_H H_i)$$

$$Y_i \sim \text{Bernoulli}(p_i)$$

where:

- (E_i) denotes the exposure proxy
- (H_i) denotes the hazard proxy
- ($Y_i \in \{0,1\}$) denotes the impact indicator

Priors

Weakly informative priors are assigned to model parameters:

$$\alpha, \beta_E, \beta_H \sim \mathcal{N}(0, 2.5)$$

Inference

Posterior inference is performed using **PyMC** with Hamiltonian Monte Carlo sampling.

Diagnostics include:

- $\hat{R}$ convergence diagnostics
- effective sample size (ESS)
- posterior predictive checks

Posterior draws produce asset-level probability distributions of impact.

We sample using NUTS in PyMC with 2 chains, 500 tuning steps and 500 posterior draws per chain (target_accept = 0.9). Citywide decision metrics are computed by applying posterior draws to the full building set ($N = 221,324$).

Posterior distribution

The posterior distribution over model parameters is

$$p(\theta \mid \text{data}) \propto p(\text{data} \mid \theta) p(\theta)$$

where

$$\theta = (\alpha, \beta_E, \beta_H)$$

5. Posterior-Derived Decision Metrics

Rather than relying on point estimates, prioritisation decisions are derived from posterior draws.

Posterior mean risk

For each asset (i):

$$p_{\text{mean},i} = \mathbb{E}_{p(\theta|\text{data})} [P(Y_i = 1 \mid \theta)]$$

The distribution of posterior mean probabilities across the city is analysed in the Results section.

Top-k membership probability

For a prioritisation threshold (k), the probability that asset (i) belongs to the top-k highest risk assets is:

$$P(i \in \text{Top}_k \mid \text{posterior})$$

Decision stability metrics

Several metrics quantify ranking stability:

- **Top-k overlap ratio**
- **Posterior entropy of membership probability**
- **Borderline share**, defined as assets with

$$0.2 < P(i \in \text{Top}_k) < 0.8$$

Computational implementation

All analyses are implemented in Python using PyMC for Bayesian inference and ArviZ for posterior diagnostics. The full analysis pipeline and figure generation scripts are provided in the accompanying repository to ensure reproducibility.

6. Results

The analysis covers **221,324 buildings** across the municipality of Rotterdam.

6.1 Posterior risk distribution

Figure 4 shows the empirical cumulative distribution of posterior mean impact probabilities across all buildings.

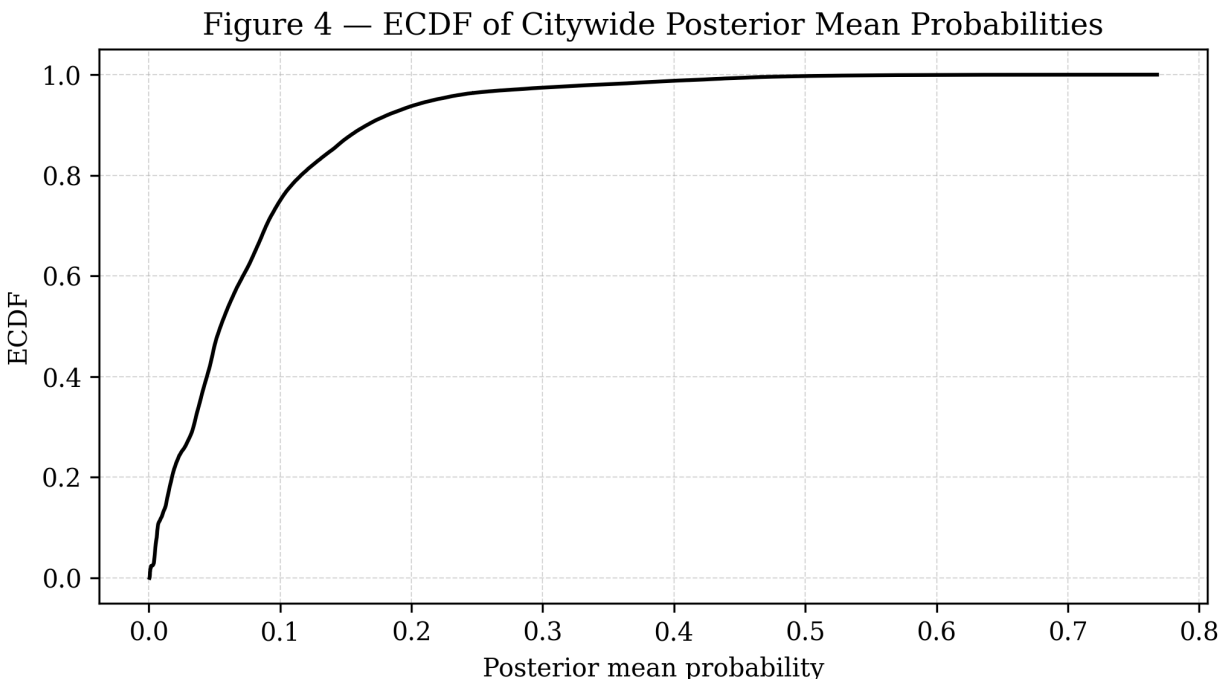


Figure 4: Empirical cumulative distribution of posterior mean probabilities

Posterior mean probabilities are strongly concentrated near low values, with a **mean of 0.077**, a **median of 0.055**, and a **90th percentile of 0.166**. Only a small fraction of assets occupy the upper tail of the distribution, with the **99th percentile reaching 0.421** and a maximum value of **0.768**.

This skewed distribution reflects the sparse structure of the synthetic outcome generation process and provides a useful setting for analysing how posterior uncertainty affects prioritisation decisions.

6.2 Decision stability across prioritisation thresholds

Decision stability is evaluated by analysing how frequently assets fall within the top-k highest risk positions across posterior draws.

The borderline share represents assets whose top-k membership probability lies between 0.2 and 0.8. These assets occupy the decision boundary where prioritisation outcomes are sensitive to posterior uncertainty.

Across prioritisation thresholds, the borderline share remains extremely small relative to the total asset population. For $k = 1000$, the borderline share is **0.053%**, increasing slightly to **0.29%** for $k = 5000$. Even at larger prioritisation sets, only a very small subset of assets exhibits unstable prioritisation behaviour.

This indicates that posterior uncertainty affects only a narrow decision boundary rather than the overall prioritisation structure.

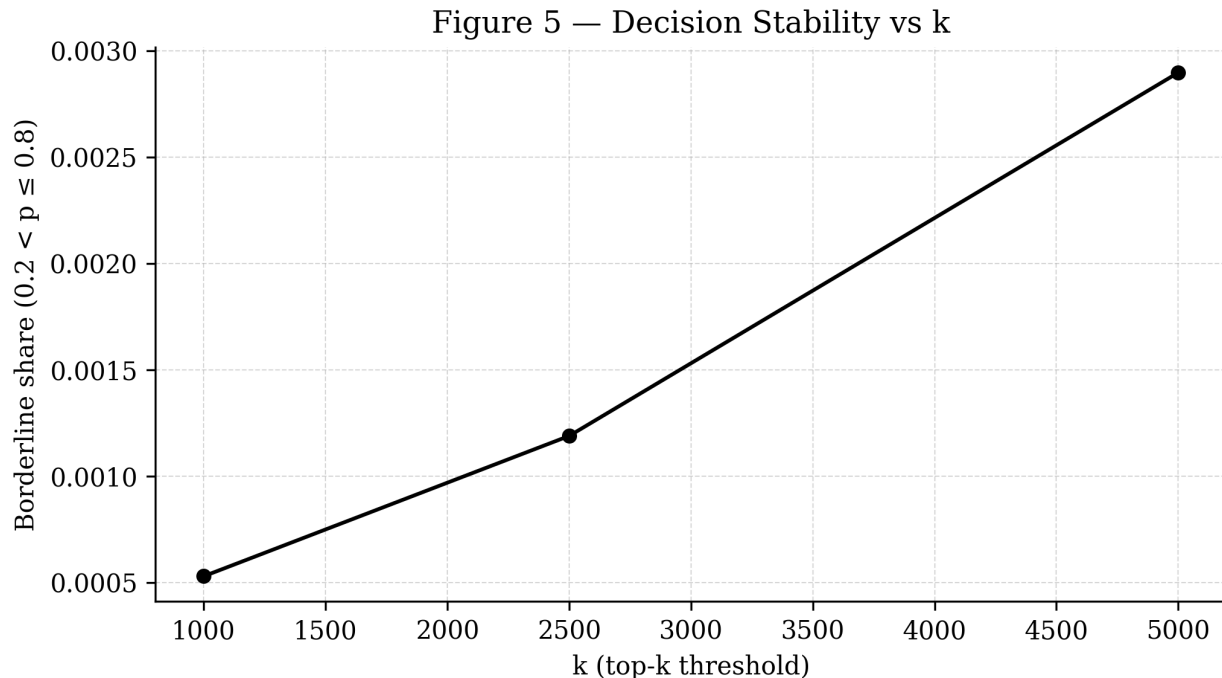


Figure 5: Borderline share across prioritisation thresholds

6.3 Borderline decision region

To better understand where instability occurs, we examine the distribution of top-k membership probabilities across assets.

For the **Top-1000 prioritisation threshold**, the mean membership probability is **0.0045**, consistent with the expected proportion of selected assets in the citywide dataset. Most assets have membership probabilities close to either **0** or **1**, indicating stable prioritisation outcomes.

Only a very small fraction of assets fall within the intermediate region ($0.2 < P(i \text{ Top}_k) < 0.8$), with a borderline share of approximately **0.00052**. These assets form a narrow decision boundary where ranking positions fluctuate across posterior draws and where additional information or improved modelling could most influence prioritisation decisions.

Overall, these results indicate that posterior uncertainty affects only a small fraction of assets near the prioritisation boundary, while the majority of ranking outcomes remain stable across posterior draws. This behaviour suggests that probabilistic prioritisation can help identify locations where additional information may most improve decision reliability.

7. Discussion

This study demonstrates how prioritisation decisions in urban risk assessment can be analysed as probabilistic objects rather than deterministic rankings. By deriving decision metrics directly from posterior distributions, the framework makes it possible to evaluate not only which assets appear most at risk, but also how stable those prioritisation decisions remain under model uncertainty.

The Rotterdam case study indicates that posterior uncertainty primarily affects a narrow subset of assets located near the prioritisation boundary. Most assets exhibit membership probabilities close to either zero or one, suggesting that the majority of ranking outcomes remain stable across posterior draws. Instability

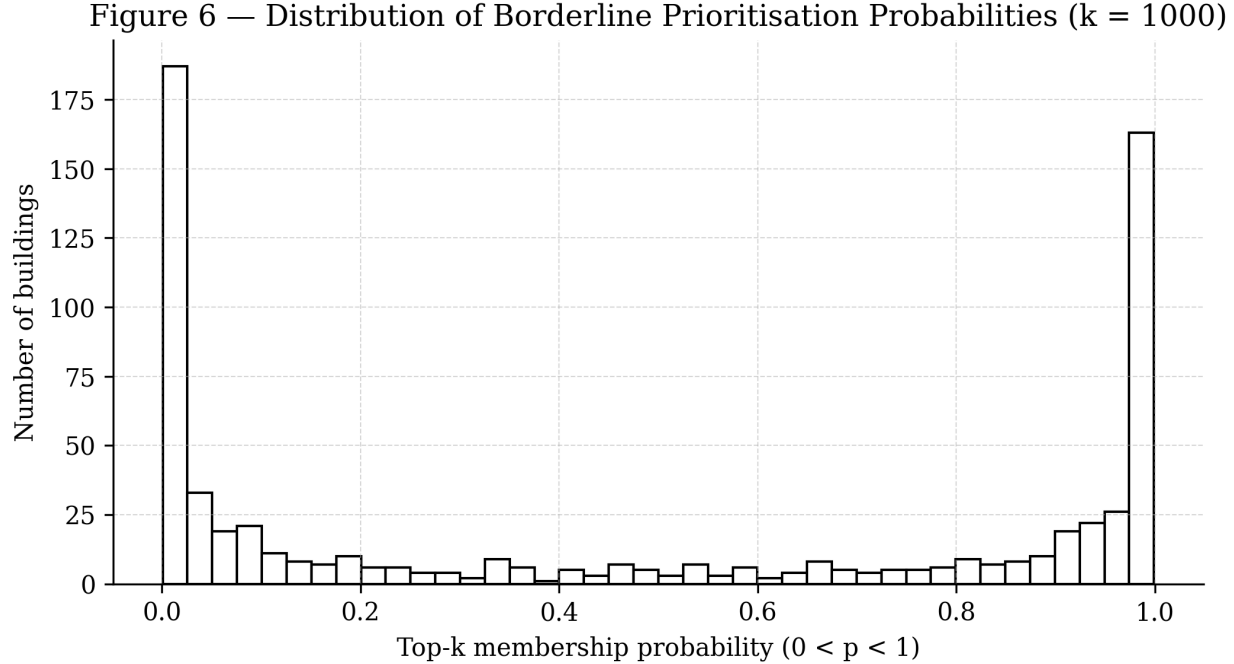


Figure 6: Distribution of top-k membership probabilities

is concentrated in a relatively small decision boundary where ranking positions fluctuate. Identifying this boundary is useful for decision makers, as it highlights locations where additional data collection or improved modelling could most influence intervention priorities.

More broadly, this work illustrates how probabilistic inference can be linked directly to decision-relevant quantities. Instead of summarising uncertainty solely through parameter estimates or predictive intervals, the approach evaluates uncertainty in the prioritisation outcome itself. This perspective allows risk assessments to distinguish between robust decisions and decisions that are sensitive to modelling assumptions. Previous work has often used probabilistic models to improve parameter estimation or predictive performance within specific components of flood risk assessment, such as damage functions or susceptibility mapping (Sairam et al., 2019; Wu et al., 2019). In contrast, the present framework evaluates uncertainty in the prioritisation outcome itself by analysing posterior ranking distributions and top-k membership probabilities.

Phase-1 intentionally employs simplified hazard and outcome representations in order to validate the statistical structure of the inference and decision pipeline. Future work will extend the framework in two directions. First, more realistic hazard representations will be introduced to account for variability in pluvial forcing and spatial flood processes. Second, the framework will be applied across multiple cities to evaluate how prioritisation stability behaves under domain shift and structural uncertainty.

Together, these extensions will allow the proposed framework to assess not only the stability of prioritisation decisions within a single urban system, but also their robustness across different environmental and modelling contexts.

8. Limitations

Several limitations should be noted:

- The hazard representation is simplified and does not model physical flood dynamics.

- The outcome variable is synthetic and not calibrated to observed damage.
- Exposure proxies are limited to hydrographic proximity indicators.

These simplifications are intentional in Phase-1 to isolate the statistical behaviour of the inference and decision pipeline.

9. Conclusion

This paper introduces a probabilistic framework for analysing prioritisation stability in urban flood risk assessment.

By deriving decision metrics directly from posterior distributions, the framework enables explicit evaluation of ranking robustness.

The Rotterdam case study demonstrates that prioritisation instability is concentrated in a small subset of assets, highlighting where additional data or modelling effort may most improve decision reliability.

Future work will extend this framework to hazard uncertainty and cross-city transfer experiments.

References

- Hall, J. W., & Harvey, H. (2009). Decision making under severe uncertainties for flood risk management: A case study of Info-Gap robustness analysis. *Journal of Flood Risk Management*. <https://doi.org/10.1111/j.1753-318X.2009.01034.x>
- Lv, H., Wu, Z., Guan, X., & Meng, Y. (2021). The construction of flood loss ratio function in cities lacking loss data based on dynamic proportional substitution and hierarchical Bayesian model. *Journal of Hydrology*. <https://doi.org/10.1016/j.jhydrol.2020.125797>
- Matrosov, E. S., et al. (2013). Robust Decision Making and Info-Gap Decision Theory for water resource system planning. *Journal of Hydrology*. <https://doi.org/10.1016/j.jhydrol.2013.03.006>
- McMillan, H., & Brasington, J. (2008). End-to-end flood risk assessment: A coupled model cascade with uncertainty estimation. *Water Resources Research*. <https://doi.org/10.1029/2007WR005995>
- Mohor, G. S., et al. (2021). Residential flood loss estimated from Bayesian multilevel models. *Natural Hazards and Earth System Sciences*. <https://doi.org/10.5194/nhess-21-1599-2021>
- Sairam, N., Schröter, K., Rözer, V., Merz, B., & Kreibich, H. (2019). A Bayesian hierarchical model for flood damage estimation in data-scarce regions. *Water Resources Research*. <https://doi.org/10.1029/2019WR025068>
- Wu, Y., et al. (2019). Assessing urban flood disaster risk using Bayesian network model and GIS applications. *International Journal of River Basin Management*. <https://doi.org/10.1080/19475705.2019.1685010>
- Wu, Y., et al. (2020). Urban flood disaster risk evaluation based on ontology and Bayesian Network. *Journal of Hydrology*. <https://doi.org/10.1016/j.jhydrol.2020.124596>